Positive Lyness Dynamics: Elliptic Translation and Rational Period Obstructions

HCS-C390 source-theorem and reproducibility package

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Abstract

We reconstruct the elliptic and arithmetic boundary of positive Lyness dynamics. Every noncentral positive orbit lies on a smooth projective cubic; a quartic discriminant separates all singular energies from the regular positive fibers. The map acts by translation by a specified rational point when the parameter and energy are rational. Consequently a rational periodic orbit forces that translation point, not its starting point, to be torsion. Mazur's theorem is an explicit deep input, and the sharper positive-coordinate period restriction to one, five and nine is credited to the established Lyness classification. Five occurs away from the center only at parameter one, while an exact parameter-seven example supplies a complete nine-cycle. The full real Abel rotation and return matrix description is retained. No rational point is presumed to exist on an arbitrary rational-energy oval, and no classification is extrapolated from a finite rational search. This source reconstruction does not supply an unbounded rational-prime periodic carrier.

中文摘要

本文进一步闭合正实莱尼斯动力学的椭圆与有理边界。 四次判别式把所有奇异能量与正则正实纤维分开；
映射在光滑射影三次曲线上表现为指定点的平移。 当参数及初值有理时，周期性迫使平移点成为有理扭点，
而不是迫使初始点本身成为扭点。 马祖尔扭点定理与正坐标周期只能为一、五、九的经典分类， 均被明确列为外部输入。
非中心五周期只发生在参数一；参数七的精确例子给出完整九周期。
本文保留全部实旋转与返回矩阵，不把有理能量误认为必有有理点， 也不从有限搜索推断普遍分类。

Keywords: Lyness map; elliptic oval; Abel rotation; rational torsion; return shear; orbit counting
关键词：莱尼斯映射；椭圆卵形线；阿贝尔旋转；有理扭点；返回剪切；轨道计数

1 One source, two periodicity questions

For $a > 0$ consider the autonomous map and its invariant

$$F_a(x,y) = \left(y, \frac{a+y}{x}\right), \quad Q = (0,\infty)^2, \quad H_a(x,y) = \frac{(x+1)(y+1)(x+y+a)}{xy}. \tag{1}$$

A least period is a positive iterate count. In particular, the historical term "prime period" can mean least period; here a *prime integer* always means an integer prime. The rational subset is considered only when $a \in \mathbb{Q}_{>0}$; it is not interchangeable with the full real quadrant. The complete-real question asks for every point's oval, rotation, repetitions and return derivative. The arithmetic question asks which of those periodic families actually contain positive rational points.

Ownership and local progress. The autonomous Lyness family has strong classical owners. Bastien–Rogalski give its global real behavior and rotation endpoints [1]. Beukers–Cushman prove an autonomous monotonicity theorem [2]; this paper does not need global monotonicity and does not turn it into a claim of a nonzero derivative everywhere. Gasull–Mañosa–Xarles study rational periods and the elliptic normal form [3]. The deep rational torsion input is Mazur's theorem [4]. The repository predecessor C173 treated only the identity $F_1^5 = I$. Here the actual step is the whole positive-real foliation and its arithmetic boundary, not another finite prefix at the same parameter. All literature-dependent conclusions below are labeled; no priority claim or target spectral identification is made.

2 Every positive orbit is accounted for

Let $\ell = (1+\sqrt{1+4a})/2$, $c_a = (\ell,\ell)$ and $h_* = (\ell+1)^3/\ell$. Put $\omega = dx \wedge dy/(xy)$ and $R(x,y) = (y,x)$. Direct substitution gives

$$F_a^{-1}(x,y) = ((a+x)/y, x), \quad RF_aR = F_a^{-1}, \quad F_a^*\omega = \omega, \quad R^*\omega = -\omega. \tag{2}$$

The Cartesian determinant is $(a+y)/x^2$, not one. Its ratio to the new coordinate product is $1/(xy)$, which proves the weighted area identity.

Theorem 1 (Global positive exhaustion). *The point c_a is the unique fixed point and the unique critical point of H_a in Q. The positive fiber at $h = h_*$ is $\{c_a\}$. Every $h > h_*$ has precisely one compact analytic oval $C^+_{a,h}$. Every noncentral two-sided orbit lies on its unique such oval, bounded away from both axes.*

Proof. In $u = \log x, v = \log y$, the invariant expands to

$$K_a = e^u + e^v + e^{u-v} + e^{v-u} + (a+1)(e^{-u} + e^{-v}) + ae^{-u-v} + a + 2. \tag{3}$$

The Hessian is a sum of positive multiples of exponent-vector outer products; the vectors $(1,0)$ and $(0,1)$ already span the plane. Thus it is positive definite. The positive and negative coordinate exponentials make K_a proper. It has a unique nondegenerate minimum; symmetry and

$$\frac{d}{dr}H_a(r,r) = \frac{2(r+1)(r^2-r-a)}{r^3}$$

locate it at c_a. Along each ray from the minimum in log coordinates, strict convexity gives a strictly increasing value after the minimum, tending to infinity. Each higher level is therefore one smooth radial graph. The inverse in (2) preserves positivity, and the invariant confines every forward and backward iterate to that compact level. Solving $F_a(x,y) = (x,y)$ gives the same unique center. □

3 A normalized Abel rotation and all return matrices

Write the level equation as

$$(x+1)y^2 + B(x)y + (x+1)(x+a) = 0, \quad B(x) = (x+1)(x+a+1) - hx, \quad D(x) = B(x)^2 - 4(x+1)^2(x+a). \tag{4}$$

Let $p_h = (r,r)$ be the unique diagonal intersection with $r > \ell$. Let α, β be the minimum and maximum x coordinates on the positive oval; they are the two turning roots delimiting that oval, not arbitrarily chosen roots of the quartic. Define

$$T(h) = 2\int_\alpha^\beta \frac{dx}{\sqrt{D(x)}}, \qquad \tau(h) = 2\int_r^\beta \frac{dx}{\sqrt{D(x)}}, \qquad \rho(h) = \frac{\tau(h)}{T(h)}. \tag{5}$$

These are positive finite Abel integrals with their positive square root on the oval interval.

Theorem 2 (Complete oval dynamics). *The function $\rho : (h_*, \infty) \to (0,1)$ is real analytic. There are smooth coordinates (h,θ) on every regular annulus such that*

$$F_a(h,\theta) = (h, \theta + \rho(h)), \qquad R(h,\theta) = (h, -\theta), \quad \theta \in \mathbb{R}/\mathbb{Z}. \tag{6}$$

If $\rho(h) = m/n$ in lowest terms, every point of that oval has least period n; otherwise every two-sided orbit is dense on the oval. For every positive integer N,

$$\mathrm{Fix}(F_a^N) = \{c_a\} \cup \bigcup_{\substack{h>h_* \\ N\rho(h)\in\mathbb{Z}}} C^+_{a,h}. \tag{7}$$

At a period-n point the return matrix, in these coordinates, is

$$DF_a^{kn} = \begin{pmatrix} 1 & 0 \\ kn\rho'(h) & 1 \end{pmatrix}, \qquad k \geq 1. \tag{8}$$

Its multipliers are both one. A nonzero shear is asserted only when $\rho'(h) \neq 0$.

Proof. The auxiliary Hamiltonian vector field $X_H = xy(H_y\partial_x - H_x\partial_y)$ satisfies $\iota_{X_H}\omega = dH$ and is nonzero on each compact regular oval. The map commutes with its complete oval flow because it preserves both H and ω. On the upper and lower quadratic roots in (4), respectively, $\dot{x} = \pm\sqrt{D}$. The full flow period is therefore T. At $x = r$, the two roots are r and $(a+r)/r < r$; the latter is exactly $F_a(p_h)$'s second coordinate. The positive clockwise arc from p_h through β to that point takes time τ. Elapsed flow time from p_h, divided by T, defines θ and proves (6). The swap fixes the section and reverses the flow. Removing the square-root turning singularities by $x = \alpha + (\beta - \alpha)\sin^2 t$ proves analytic dependence on each compact regular energy interval. The turning points are simple; equivalently their affine gradient is nonzero and the log-convex oval has a nondegenerate vertical tangency. The rotation on the circle then proves the period and density statements. Iteration and differentiation of (6) give (7) and (8). □

The two-by-two Cartesian monodromy at a return point is conjugate to (8); its nilpotent part squares to zero. At the center the one-step characteristic polynomial is $\lambda^2 - \ell^{-1}\lambda + 1$, so the multipliers are $\exp(\pm 2\pi i \rho_0)$, where

$$\rho_0(a) = \frac{1}{2\pi} \arccos \frac{1}{2\ell}. \tag{9}$$

The center is Lyapunov stable by its invariant minimum, not asymptotically stable. Periodic ovals are continuous families, not isolated hyperbolic orbits. The auxiliary T and τ have not altered the source's unit-iterate clock.

4 Smooth cubics, exceptional parameters and rational torsion

The projective level curve is

$$P_{a,h}(X,Y,Z) = (X+Z)(Y+Z)(X+Y+aZ) - hXYZ = 0. \tag{10}$$

The exact quartic elimination gives

$$\operatorname{disc}_x D = 256h^3(a-h-1)^2 a(h-h_+)(h-h_-), \qquad h_\pm = \frac{2a^2+10a-1 \pm (4a+1)\sqrt{4a+1}}{2a}. \tag{11}$$

Here $h_+ = h_*$. Since $a = \ell^2 - \ell$, $h_* - a - 2 = 4\ell + 1 + 1/\ell > 0$. The possible singular energies $0, a-1, h_\pm$ do not meet any regular positive oval. More explicitly, for $x \neq -1$ an affine singular point would force a repeated root of D; at $x = -1$ the curve forces $y = 0$ and has $P_y = h \neq 0$. The three points at infinity each have a nonzero partial derivative. Thus every $h > h_*$ gives a smooth complex genus-one cubic. The singular center fiber is not subjected to an elliptic torsion theorem.

Proposition 1 (Translation point, not starting-point torsion)**.** *With identity $O = [1 : -1 : 0]$, the smooth cubic's group law gives $F_a(P) = P + Q_0$, where $Q_0 = [1 : 0 : 0]$. If a, x, y are positive rational and $P = (x,y)$ is noncentral, its finite least period is the order of Q_0 in this rational elliptic group.*

Proof. The tangent at O is $X + Y = (h-a)Z$; substitution gives $h(h-a+1)Z^3$, so O is a flex. The swap is negation. The involution $J(x,y) = ((a+y)/x, y)$ is the sheet exchange of the degree-two projection to the y line. It has branch fixed points; the genus-one involution law therefore gives $J(P) = J(O) - P$. Its homogeneous expression $[Z(Y+aZ) : XY : XZ]$ gives $J(O) = [0 : 1 : 0] = -Q_0$. Since $F_a = R \circ J$, the translation formula follows, in agreement with [3, Section 2.1]. Rationality gives $h \in \mathbb{Q}$ and a smooth curve over $\mathbb{Q}$, so $F_a^n(P) = P$ is equivalent to $nQ_0 = O$. No assertion that P itself is torsion follows from this cancellation. ☐

Deep and sharper external inputs. Mazur's torsion theorem [4] limits finite point orders on a rational elliptic curve to $1, \ldots, 10, 12$. It is not proved by this package. The additional positive-coordinate exclusion, credited to the Lyness classification discussed in [3], sharpens positive rational least periods to

$$\{1, 5, 9\}. \tag{12}$$

This sharper restriction is explicitly imported, not a consequence of Mazur's list alone and not an inference from our finite grid. A positive rational center exists exactly when $a = r(r-1)$ for a rational $r > 1$, with center (r,r).

The five-periodic parameter and an exact nine-cycle. Direct iteration gives $F_1^5 = I$; all noncentral positive points then have least period five. Conversely, for $a \neq 1$, the equations $F_a^5(x,y) = (x,y)$, after positive denominators are cleared, reduce to

$$\begin{aligned} p &= ax - ay + a - y^2 + y = 0, \\ q &= -a^2x - a^2 + ax^2y + ax^2 - axy + ax - 2ay + xy - y^2 = 0. \end{aligned}$$

Substitute $x = (a+y)(y-1)/a$ from $p = 0$ into q to obtain $q = y(a+y)^2(y^2 - y - a)/a$. Positivity forces $x = y = \ell$. There is therefore no noncentral period five at $a \neq 1$. At $a = 7$, the successive scalar coordinates

$$3/2, \quad 5/7, \quad 36/7, \quad 17, \quad 14/3, \quad 35/51, \quad 28/17, \quad 63/5, \quad 119/10 \tag{13}$$

form the GMX nine-cycle [3]. Adjacent cyclic pairs satisfy the recurrence exactly and are distinct, proving least period nine; the energy is $258/7$. This example does not establish nine-period existence for every rational parameter. Rational energy alone does not ensure a rational point on the positive oval. The excluded parameter $a = 0$ is globally six-periodic, and is not covered by a generic $a > 0, a \neq 1$ argument.

5 Evidence, dependency boundaries and conclusion

The canonical finite evidence contains 24 upper-diagonal orbit controls with 288 exact map steps, ten distinct seed-induced five-cycles and one nine-cycle, totaling 59 cycle points. Five rational centers are checked separately. Five certified intervals inside the gap between classical rotation endpoints use a 128-bit rational enclosure of π and 96-bit cosine enclosures. Their 1,280 prime and composite denominator controls give 690 reduced-fraction sufficient-period witnesses. Those witnesses are rotation values in the guaranteed image, not claimed exact coordinates of reconstructed periodic ovals.

An independent checker differentiates the scalar recurrence in two dual directions, rather than importing the producer's matrix product. Symbolic checks cover the invariant, weighted symplectic law, quartic discriminant, flex, exceptional period and return matrices. The separate 90-digit quadratures are numerical controls, not rigorous interval certificates for Abel integrals. Universal geometry and periodicity follow from the proof and the named classical inputs; finite counts do not prove them. Repaired-hash mutations test semantic rejection as well as type and YAML closure.

The source is explicitly owner-heavy. Its real oval ledger is complete, but continuous orbit families and rational torsion do not furnish a discrete rational-prime carrier. The strict evaluator tuple is
`A0_WEAK_ARITHMETIC_RELATION`, `A1_WEAK`, `A2_FAIL`, `A3_FAIL`, `A4_FORMAL_HINT`;
the overall Route-A verdict is rejected. No target local data, Euler factors, root numbers, automorphy, target zeros, divisor, counting law, functional equation or Hilbert–Pólya operator is claimed. `NO_BAD_EULER_OR_ROOT_NUMBER`. Route B remains disabled.

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Round one: smooth elliptic fibers and the rational torsion restriction.